

PHYS 143 – Mechanics and Thermal Physics

Lecture 7: Impulse, Momentum and Collisions – Tutorial Sheet

Hope Donglo, Office 24, Department of Physics, University of Ghana

Use the Master Procedure: identify, sketch before and after, check that the system is isolated, predict, select the model, execute by components, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A head-on collision of a car and a truck

A 1000 kg car moving east at 15 m/s collides head on with a 3000 kg truck moving west at 5.0 m/s. They lock together. Find the velocity after the collision and the kinetic energy lost.

Solution. Take east as $+x$: $(1000)(15) + (3000)(-5.0) = (4000)v \Rightarrow v = 0$. Both stop. $K_{\text{before}} = \frac{1}{2}(1000)(15)^2 + \frac{1}{2}(3000)(5.0)^2 = 112.5 + 37.5 = 150 \text{ kJ}$; $K_{\text{after}} = 0$, so all 150 kJ is lost. *Check:* equal and opposite momenta (15 000 kg m/s each), so the total is zero.

A2. A two-dimensional completely inelastic collision

A 2.0 kg object moving east at 4.0 m/s collides with and sticks to a 3.0 kg object moving north at 2.0 m/s. Find the velocity after the collision.

Solution. x : $(2.0)(4.0) = (5.0)v_x \Rightarrow v_x = 1.6 \text{ m/s}$. y : $(3.0)(2.0) = (5.0)v_y \Rightarrow v_y = 1.2 \text{ m/s}$. Speed 2.0 m/s at $\tan^{-1}(1.2/1.6) = 37^\circ$ north of east. $K_{\text{before}} = 16 + 6 = 22 \text{ J}$, $K_{\text{after}} = \frac{1}{2}(5.0)(2.0)^2 = 10 \text{ J}$: inelastic.

B. Practice

B1. Impulse from a graph

The force on a 0.50 kg object initially at rest rises linearly from 0 to 100 N in 0.020 s, stays at 100 N for 0.030 s, then drops to zero. Find the impulse and the final speed.

B2. A bouncing ball

A 0.20 kg ball dropped from 2.0 m rebounds to 1.2 m. The contact with the floor lasts 10 ms. Find the impulse, the average force and the coefficient of restitution.

B3. Recoil

A 70 kg skater at rest on frictionless ice throws a 2.0 kg ball horizontally at 12 m/s. Find the skater's velocity.

B4. An elastic collision

A 2.0 kg block moving at 5.0 m/s collides elastically with a 3.0 kg block at rest. Find both velocities after the collision, and verify that kinetic energy is conserved.

B5. A glancing collision

A 1.0 kg ball moving east at 6.0 m/s hits a 2.0 kg ball at rest. The 1.0 kg ball leaves at 3.0 m/s, 60° north of east. Find the velocity of the 2.0 kg ball, and classify the collision.

B6. A ballistic pendulum

A 50 g bullet embeds in a 4.0 kg pendulum block, which rises 0.15 m. Find the bullet's speed.

B7. Challenge: the method is not given

A 0.15 kg ball moving at 10 m/s hits a 1.0 kg block at rest on a rough floor ($\mu_k = 0.30$), and rebounds at 4.0 m/s. How far does the block slide? What is the coefficient of restitution?

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

Items E1 and E2 share this information: A 0.15 kg ball strikes a wall at 25.0 m/s perpendicular to it and rebounds along the same line at 20.0 m/s. The contact lasts 0.012 s.

E1. The magnitude of the impulse the wall exerts on the ball is

- (A) 0.75 N s
- (B) 3.00 N s
- (C) 3.75 N s
- (D) 6.75 N s
- (E) 7.50 N s

E2. The magnitude of the average force of the wall on the ball is

- (A) 0.0810 N (B) 1.47 N (C) 6.75 N (D) 281 N (E) 562 N

E3. Which of the following statements is/are true?

- I. In a perfectly inelastic collision the total momentum is conserved but kinetic energy is not.
- II. In a one-dimensional elastic collision the relative speed of approach equals the relative speed of separation.
- III. In any collision of an isolated system the total kinetic energy is conserved.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

Items E4 and E5 share this information: A 1500 kg car moving at 12.0 m/s collides head-on with a stationary 2500 kg truck. The two vehicles lock together.

E4. The common speed immediately after the collision is

- (A) 4.5 m/s (B) 6.0 m/s (C) 7.2 m/s (D) 7.5 m/s (E) 32.0 m/s

E5. The fraction of the initial kinetic energy that is transformed into other forms is

- (A) 0.250 (B) 0.375 (C) 0.500 (D) 0.625 (E) 1.000

E6. A 2.0 kg cart moving at 6.0 m/s makes an elastic head-on collision with a 4.0 kg cart at rest. The velocity of the 4.0 kg cart afterwards is

- (A) -2.0 m/s (B) 2.0 m/s (C) 3.0 m/s (D) 4.0 m/s (E) 6.0 m/s

E7. A 60.0 kg skater at rest on smooth ice throws a 0.50 kg ball horizontally at 20.0 m/s. The speed of recoil of the skater is

- (A) 0.167 m/s
(B) 0.333 m/s
(C) 10 m/s
(D) 20 m/s
(E) 2400 m/s

E8. A 1000 kg car travelling east at 20.0 m/s collides at a junction with a 1500 kg van travelling north at 12.0 m/s, and they move together. The speed of the wreckage just after the collision is

- (A) 7.2 m/s
(B) 8.0 m/s
(C) 10.8 m/s
(D) 15.2 m/s
(E) 16.0 m/s

E9. A 10 g bullet embeds in a 1.99 kg block hanging on light strings, and the block swings up by 5.0 cm. The speed of the bullet just before impact was

- (A) 0.99 m/s (B) 1.98 m/s (C) 198 m/s (D) 99 m/s (E) 396 m/s

E10. Two objects A and B have the same kinetic energy, and the mass of A is four times the mass of B. The ratio of the momentum of A to that of B, p_A/p_B , is

- (A) 0.25 (B) 2.00 (C) 0.50 (D) 1.00 (E) 4.00

Answers

- B1 $J = 4.0 \text{ N s}$; $v = 8.0 \text{ m/s}$
B2 $J = 2.2 \text{ N s}$ upward; $F_{\text{av}} = 2.2 \times 10^2 \text{ N}$; $e = \sqrt{1.2/2.0} = 0.77$
B3 -0.34 m/s (opposite to the ball)
B4 $v'_1 = -1.0 \text{ m/s}$, $v'_2 = +4.0 \text{ m/s}$; $K = 25 \text{ J}$ before and after
B5 2.6 m/s at 30° south of east; inelastic ($K: 18 \text{ J} \rightarrow 11 \text{ J}$)
B6 $v = \frac{(M+m)\sqrt{2gh}}{m} = 1.4 \times 10^2 \text{ m/s}$
B7 block speed 2.1 m/s ; slides 0.75 m ; $e = 0.61$

Exam-style practice: answer key and brief solutions

Key: E1 D E2 E E3 B E4 A E5 D E6 D E7 A E8 C E9 C E10 B

E1 (D). Take $+x$ away from the wall: $p_{ix} = -m(25.0) = -3.75$ and $p_{fx} = +m(20.0) = +3.00 \text{ kg m/s}$. $J = \Delta p = 3.00 - (-3.75) = 6.75 \text{ N s}$. The velocity reverses, so the speeds add: using $m(v_1 - v_2) = 0.75$ is the usual slip.

E2 (E). $J = F_{\text{av}}\Delta t \Rightarrow F_{\text{av}} = \frac{6.75}{0.012} = 562 \text{ N}$, using the impulse from the previous item. This is about 380 times the ball's weight, which is why the weight can be neglected during the contact.

E3 (B). I is true: the colliding bodies move together afterwards and some K becomes internal energy. II is true for elastic collisions. III is false: only in elastic collisions. Momentum is conserved in all collisions of an isolated system. Answer: I and II only.

E4 (A). Momentum is conserved (the system is the car and truck; the collision forces are internal): $m_1v_1 = (m_1 + m_2)v_f$, so $v_f = \frac{(1500)(12.0)}{4000} = 4.5 \text{ m/s}$. Check: $v_f < v_1$, in the same direction as the car's initial velocity.

E5 (D). $K_i = \frac{1}{2}(1500)(12.0)^2 = 1.08 \times 10^5 \text{ J}$ and $K_f = \frac{1}{2}(4000)(4.5)^2 = 4.05 \times 10^4 \text{ J}$, so the fraction lost is $1 - K_f/K_i = 0.625$, which equals $m_2/(m_1 + m_2)$. It is not zero: the collision is not elastic.

E6 (D). Elastic and one-dimensional: $v_{2f} = \frac{2m_1}{m_1 + m_2}u = \frac{4.0}{6.0}(6.0) = 4.0 \text{ m/s}$ in the original direction (and $v_{1f} = -2.0 \text{ m/s}$). Check momentum: $12.0 = 2.0(-2.0) + 4.0(4.0)$ and kinetic energy: $36 = 4 + 32 \text{ (J)}$. ✓

E7 (A). The initial momentum is zero and is conserved: $0 = Mv_r + m_bv_b$, so $v_r = \frac{m_bv_b}{M} = \frac{(0.50)(20.0)}{60.0} = 0.167 \text{ m/s}$, opposite to the ball. Check: $v_r \ll v_b$ because $M \gg m_b$.

E8 (C). Conserve each component of momentum: $x: 1000(20.0) = 2500v_x \Rightarrow v_x = 8.0$; $y: 1500(12.0) = 2500v_y \Rightarrow v_y = 7.2 \text{ m/s}$. Speed $= \sqrt{8.0^2 + 7.2^2} = 10.8 \text{ m/s}$, directed 42° north of east (sketch: between the two initial directions, closer to east). The scalar sum $(20.0 \cdot 1000 + 12.0 \cdot 1500)/2500 = 15.2$ is wrong because momenta are vectors.

E9 (C). Two stages. Swing (energy): $\frac{1}{2}(m + M)V^2 = (m + M)gh \Rightarrow V = \sqrt{2gh} = 0.99 \text{ m/s}$. Collision (momentum): $mv = (m + M)V \Rightarrow v = \frac{2.00}{0.010}(0.99) = 198 \text{ m/s}$. Energy is *not* conserved in the collision stage, so one equation cannot do both.

E10 (B). $p = \sqrt{2mK}$, so for equal $K: p_A/p_B = \sqrt{m_A/m_B} = \sqrt{4} = 2$. Check with numbers: $m_B =$

1, $K = 8$; $p_B = 4$; $m_A = 4$; $p_A = \sqrt{2(4)(8)} = 8$. The ratio of speeds is $1/2$, but the momentum ratio is $m_A v_A / m_B v_B = 4 \times \frac{1}{2} = 2$.